

Python Cheatsheet

MATH60033A · Session 07 · PCA and factor analysis

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Everything you need to type, in one place. The first three sections are the same every week; the last one is the Python from session 6.

Starting Python in VS Codium

Open the course folder once — **File > Open Folder**, then pick the repository. Everything below assumes you are inside it.

Open a terminal with **View > Terminal** (Ctrl + ` on Windows and Linux, Cmd + ` on macOS). Create the environment the first time only:

```
python3 -m venv .venv      # macOS
python3 -m venv .venv      # Linux
py -m venv .venv           # Windows (PowerShell)
```

Then activate it — **every time you open a new terminal**:

```
source .venv/bin/activate  # macOS
source .venv/bin/activate  # Linux
.venv\Scripts\activate     # Windows (PowerShell)
```

You will see (.venv) at the start of the prompt. If you do not, nothing below will find pandas. Install the packages once, with the environment active:

```
pip install -r requirements.txt
```

Finally tell the editor which Python to use: Ctrl/Cmd + Shift + P, type **Python: Select Interpreter**, and choose the one inside .venv.

A new file, and running it

File > New File, then save it with a .py ending — session03.py, say. Write your code, save it, and in the terminal:

```
python session03.py        # macOS, Linux and Windows alike, inside .venv
```

Inside the environment the command is **python** on all three platforms. Outside it, macOS and Linux need **python3** and Windows needs **py**, which is one more reason to activate first.

For trying one line at a time, type **python** on its own to get the >>> prompt, and **exit()** to leave. The editor's **Run** triangle does the same thing as the command above.

The four things you always do

```
import pandas as pd
```

```
# 1 - build a DataFrame
df = pd.DataFrame({"country": ["FR", "DE", "IT"],
                   "gdp_pc_eur": [37_000, 46_000, 32_000]})
```

```
# 2 - look at what is in it
df.head()      # the first five rows
df.shape       # (rows, columns)
df.dtypes      # what type each column holds
```

```

df.info()          # types, and how many values are missing
df.describe()      # count, mean, sd and quartiles, per numeric column

# 3 - read a file in
df = pd.read_csv("data.csv")
df = pd.read_excel("data.xlsx")          # needs openpyxl
df = pd.read_parquet("data/spine/core.parquet")

import qmib
df = qmib.load("core")                   # a course dataset, by name

# 4 - write a CSV out
df.to_csv("out.csv", index=False)      # index=False, or you get a stray column

```

From session 6

Fit

```

from linearmodels.panel import PanelOLS
fit = PanelOLS.from_formula("y ~ x + EntityEffects", data=panel).fit()

RandomEffects.from_formula("y ~ 1 + x", data=panel).fit()

smf.ols("price ~ living_area + I(living_area**2)", data=d).fit()

smf.ols("wage ~ age * female", data=d).fit()
# a * b expands to a + b + a:b

```

Choose and validate

```

big.compare_f_test(small)          # nested models only

np.linalg.pinv(v_fe - v_re)        # for the Hausman statistic

```